- Each REST controller method handles an incoming request with a specific method (`GET`, `POST`, `PUT`, `DELETE`) and endpoint (e.g., `/books`).
- Controller methods can declare which endpoints and operations they handle through the annotations `@GetMapping`, `@PostMapping`, `@PutMapping`, `@DeleteMapping`, and `@RequestMapping`.
- Methods of a `@RestController` class can validate the HTTP request body before the processing happens by applying the `@Valid` annotation.
- The validation constraints for a given Java object are defined using annotations from the Java Bean Validation API on the fields (for example, `@NotBlank`, `@Pattern`, `@Positive`).
- Java exceptions thrown during the processing of an HTTP request can be mapped to an HTTP status code and body in a centralized `@RestControllerAdvice` class, decoupling the exception handling for the REST API from the code throwing the exception.
- Unit tests are not aware of the Spring configuration but can be written as standard Java tests using familiar tools like JUnit, Mockito, and AssertJ.
- Integration tests need a Spring application context to run. A full application context, including an optional embedded server, can be initialized for testing using the `@SpringBootTest` annotation.